# Exit Ticket · reflect and retrieve

## Exit Ticket

A program written using fewer assembly instructions always executes faster.

## Use key vocabulary from memory

1. Explain RISC in your own words.

2. Explain CISC in your own words.

3. Explain load/store in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## RISC and CISC

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why can instruction fetch and data access contend in Von Neumann architecture?

### Q2

What is a pipeline hazard?

### Q3

What is an opcode?

### Q4

Array [2,4,6,8,10,12,14,16,18], target 16. Give midpoint indexes inspected with low=0 high=8 and integer midpoint.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

RISC approach — Typically favours a smaller set of comparatively simple, regular instructions, often using explicit load/store operations.

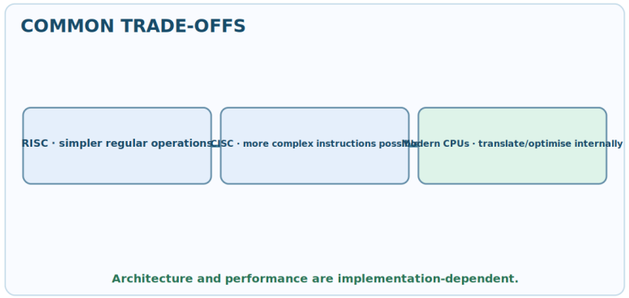
## Core idea 2

CISC approach — Can offer complex instructions that express several low-level operations in one instruction, potentially improving code density.

## Core idea 3

Avoid oversimplifying — Fewer instructions in assembly do not guarantee fewer clock cycles. Actual performance depends on pipelining, decoding, caching and processor design.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Evaluate the claim 'RISC is always faster because its instructions are simpler'. Include code density, decoding and pipelining.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

State two characteristics of RISC and two of CISC.

## Question 2

Compare one CISC-style memory-add instruction against a RISC-style LOAD, LOAD, ADD, STORE sequence for the same task.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

RISC — Reduced Instruction Set Computer design approach.

CISC — Complex Instruction Set Computer design approach.

load/store — Access memory through dedicated instructions.

code density — Amount of program function per unit of machine-code storage.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

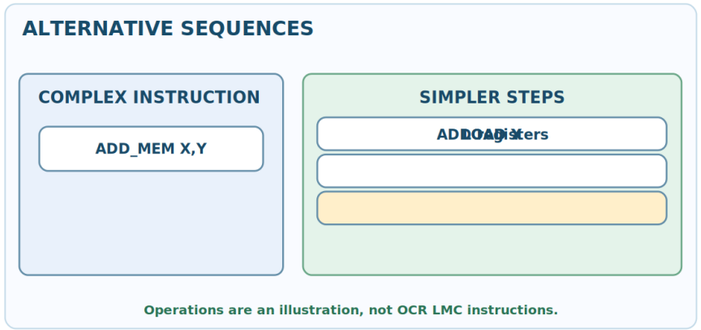
Explain why a load/store architecture needs separate load and add instructions.

## Follow the worked method

1. Predict first — Values are brought from memory into registers before the arithmetic instruction.

2. Explain and check — A regular load/store instruction set simplifies certain datapaths but may need more assembly instructions.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
The complex ISA version uses fewer listed instructions but could require more work per instruction.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.